

Cation-Infused Bilayer Ionomer Coating Enables High Partial Current Density Toward Multi Carbon Products in CO₂ Electrolysis

Mohsina Rashid, Shariful Kibria Nabil, Muflih A. Adnan, Karthick Kannimuthu, and Md Golam Kibria*

Electrochemical CO₂ reduction (eCO₂R) stands as a pivotal technology for carbon recycling by converting CO₂ into value-added products. While significant strides have been made in generating multi-carbon (C₂₊) products like ethylene (C₂H₄) and ethanol (C₂H₅OH) at industrial-scale current densities with high Faradaic efficiency (FE), cathode flooding and (bi)carbonate salt accumulation remain a fundamental concern in an alkaline electrolyte. In this work, ion-conducting polymers are used to tailor the micro-environment mitigating cathode flooding and salt precipitation and thus, enhancing the local CO₂ availability. **The impact of cation and anion exchange ionomer layers, specifically Nafion and Sustainion XA-9 are examined on overall eCO₂R performance. The use of an ultra-thin bilayer configuration significantly reduces cathode flooding and salt accumulation by ≈58% compared to commercial anion exchange membrane (AEM).** Alongside, cation infusion improves the C–C bond formation inducing a favorable micro-environment for selective C₂₊ formation. This cation-infused bilayer ionomer (CIBLI) achieves a high partial current density of ≈284 mA cm^{−2} toward C₂₊ products maintaining a stable eCO₂R performance for 24 hours (h). This scalable approach of directly deposited ultra-thin CIBLI offers a minimal conversion energy of 117 GJ/ton C₂₊ products with an energy efficiency (EE) of 29% at 350 mA cm^{−2} current density in one-step CO₂ conversion.

1. Introduction

In the realm of carbon neutrality, electrochemical CO₂ reduction (eCO₂R) stands as an advanced technology enabling the transformation of carbon dioxide (CO₂) into valuable products.^[1,2] While notable advancements have been accomplished in the production of desired multi-carbon (C₂₊) products like ethylene (C₂H₄) and ethanol (C₂H₅OH), challenges remain to achieve such feat at industrial-scale matrices including current density and Faradaic efficiency (FE) or selectivity.^[3–5] The strategies to achieve this goal involve enhancing local CO₂ concentration and avoiding unwanted as well as competing side reactions including (bi)carbonate formation and hydrogen evolution reaction (HER).^[6–8]

In this context, eCO₂R has transitioned from H-cell to membrane electrode assembly (MEA) where a gas diffusion electrode (GDE) can overcome the solubility limit of aqueous CO₂.^[9] MEA is usually accompanied by an anion exchange membrane (AEM) which can maintain a favorable micro-environment at the cathode surface and improve the eCO₂R activity.^[10–12] As far as the catalyst

is concerned, the thermodynamic potential and selectivity in eCO₂R typically hinge on the binding affinity between the metal catalyst and adsorbed CO₂ or carbon monoxide (CO).^[13] Copper (Cu) stands out among other metals due to its distinctive electronic structure, affording a moderate adsorption capacity that facilitates the generation of C₂₊ products.^[13] The effectiveness of Cu-based catalysts is also influenced by factors like morphology, surface state, defects, crystal phases, and others.^[14,15] Combining this with MEA and operating with alkaline (i.e., high pH) anolyte further enhances the selectivity toward C₂₊ products (i.e., C₂H₄) with deceleration of HER kinetics.^[16] In addition, alkaline pH allows the use of highly active and stable non-precious transition metals (i.e., Nickel) as anode catalysts with accelerated oxygen evolution reaction (OER) kinetics which reduces the overall cell voltage in eCO₂R.^[17,18] However, recent studies showed that alkaline anolyte transports through an AEM toward the

M. Rashid, S. K. Nabil, K. Kannimuthu, M. G. Kibria
Department of Chemical and Petroleum Engineering
University of Calgary
2500 University Drive, NW, Calgary, AB T2N 1N4, Canada
E-mail: md.kibria@ucalgary.ca

M. A. Adnan
Department of Chemical Engineering
Islamic University of Indonesia
Sleman Daerah Istimewa, Yogyakarta 55584, Indonesia

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/aenm.202400570>

© 2024 The Author(s). Advanced Energy Materials published by Wiley-VCH GmbH. This is an open access article under the terms of the [Creative Commons Attribution](#) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

DOI: 10.1002/aenm.202400570

cathode, resulting in (bi)carbonate salt formation and precipitation, blocking the pores of the gas diffusion layer (GDL).^[19,20] Often more CO₂ is consumed by these undesired chemical reactions other than the desired eCO₂R and thus hindering eCO₂R performance.^[21,22] To mitigate this, we have recently reported a facile approach to directly deposit an ultra-thin layer of perfluoro-sulphonic acid (PFSA) polymer (i.e., Nafion).^[23] This strategy can replace commercially available standalone membrane and systematically control ion migration between anode and cathode. However, the study was still limited in terms of the partial current density, j ($\approx 158 \text{ mA cm}^{-2}$) toward C₂₊ products which can be attributed to the reduced CO₂ availability near the catalyst vicinity.^[23]

Ion-conducting polymers or ionomers are well-known compounds with high CO₂ affinity within the organic layer. They can tailor the local reaction environment by controlling the concentrations of CO₂, water (H₂O), hydroxide (OH⁻), and proton (H⁺) due to the presence of charged hydrophilic side chains spread over a hydrophobic backbone.^[24] Ionomers are exploited for these hydrophobic and hydrophilic functionalities, where the conformally differentiated domains can favor both gas and ion transport over the catalyst surface.^[25] The hydrophobic moieties extend gas diffusion while hydrophilic domains lead to better wetability and ion transport. As a result, the reaction interface involving gas, ion, and electron, can be increased from the sub-micrometer regime to several micrometer length scale.^[24,26] Although Nafion is widely used as an ionomer for eCO₂R due to its robustness and proton conductivity, the effect of different thin film ionomers on eCO₂R is still not well devised and mostly relies on factual understanding. It is thus imperative to discern a systematic understanding of how different ionomer layers with varying properties (acidity versus alkalinity, CO₂ availability, CO₂ permeability, water uptake, ion transport, etc.) can impact eCO₂R by modulating the local micro-environment necessary for achieving high j toward C₂₊ products ($j_{\text{C}_{2+}}$).

An alternative strategy to induce high $j_{\text{C}_{2+}}$ is to employ an anion exchange ionomer like Sustainion using a direct deposition approach. Specifically Sustainion XA-9 is an alkaline ionomer with a hydrophilic (poly 4-vinylbenzyl alkyl-imidazolium chloride) unit within the nanopore network.^[27] It has been previously shown that Sustainion XA-9 can maintain ionic conductivity at the lowest specific resistance.^[28] Moreover, Sustainion XA-9 can also maintain an alkaline pH with high CO₂ solubility (20 times higher than Nafion) due to strong affinity by imidazolium groups.^[24,9,29] We hypothesize that the aforesaid properties can allow Sustainion XA-9 to provide higher CO₂ availability at the catalyst vicinity, which is the crucial factor to regulate high $j_{\text{C}_{2+}}$ while maintaining selectivity due to alkaline nature. However, hydrophilic ionomers are presumed to be filled by electrolyte during operation and the gas phase CO₂ can only diffuse in dissolved form which increases the chance of carbonation. Similar phenomenon was also observed by Kim et al. who later employed a combination of both anion and cation exchange ionomers to regulate the water concentration (CO₂/H₂O ratio) on cathode surface.^[24] Inspired by this, we hypothesized that the high water uptake and in turn the loss of CO₂ for our directly deposited membrane can be avoided by changing the surface wetability. This can also control cation electromigration as well as keep local reaction environment to be alkaline. We dictated that

by implementing a hydrophilicity gradient (coating hydrophilic Sustainion XA-9 on top of bulk hydrophobic Nafion) to induce higher CO₂ availability required for very high $j_{\text{C}_{2+}}$.

Herein, we demonstrate the effect of different ionomer layers to induce beneficial micro-environments for selective C₂₊ production with high $j_{\text{C}_{2+}}$ on sputtered Cu catalyst. We study the impact of single-layer ionomer (SLI), bilayer ionomer (BLI), and cation-infused bilayer ionomer (CIBLI) thin films on efficient eCO₂R. First, we analyzed the performance of the Sustainion XA-9 as SLI yielding a $j_{\text{C}_{2+}}$ of 280 mA cm⁻². However, it also contributed to high (bi)carbonate formation and thus high CO₂ loss because of the elevated CO₂ permeability. Later, to address these issues, we added one additional layer of Nafion with 1 M K⁺ infusion (CIBLI-1 M) which resulted in comparable $j_{\text{C}_{2+}}$ (i.e., 284 mA cm⁻²) with SLI, mostly toward C₂H₄, C₂H₅OH and acetate (C₂H₃O⁻) at a current density of 350 mA cm⁻². Hence, our CIBLI can offer a stable (24 h) eCO₂R system with significantly higher $j_{\text{C}_{2+}}$, notably suppressing the cathode flooding as well as HER and retaining the (bi)carbonate formation as AEM. Our further analysis reveals that these results were achieved due to the interdependent effects of i) optimized water diffusion through ultra-thin ionomer layers avoiding water accumulation and salt precipitation at the cathode; ii) high CO₂ availability at the cathode and iii) alkaline local environment because of the extended contact between active catalyst and ionomer layers.

2. Results and Discussion

We initiated our analysis by using commercial AEM in the MEA system to check the performance of traditional eCO₂R. As a reference for AEM, we have used Sustainion X37- 50 Grade RT membrane having the least area-specific resistance, that is, high ionic conductivity compared to other commercially available membranes in alkaline electrolyte.^[30] The MEA also consisted of a porous polytetrafluoroethylene (PTFE) as the GDL, with a 300 nanometer (nm) thick sputtered copper (Cu) as cathode catalyst and Nickel (Ni) foam as the anode (see Sections A–C, Supporting Information for details). At the cathode, humidified CO₂ was fed and at the anode, 1 M potassium hydroxide (KOH) was re-circulated as anolyte. The system schematic is shown in Figure 1a.

The j and the voltage at different current density (I) for this configuration have been demonstrated in Figure 1b. The maximum j obtained toward C₂H₄ ($j_{\text{C}_{2\text{H}_4}}$) was 55 mA cm⁻² at a current density of 150 mA cm⁻² which is consistent with the previous reports.^[31,32] The Faradaic efficiency of C₂H₄ (FE_{C₂H₄}) started to deteriorate at higher overpotentials due to an increase in HER. The total gas products distribution (Figure S3, Supporting Information) aligns with previous reports^[31,33] and reflects that there are no significant liquid products at higher currents and thus, liquid products were not analyzed further. The mass transport (at 50 mA cm⁻²) in Figure 1c indicated the excessive cation transport along with KOH from the anode to the cathode via water diffusion which is due to the high water uptake (>80 wt%)^[34] of thick (≈ 50 micrometer or μm) standalone AEM.^[35] The concentration of OH⁻, bicarbonate (HCO₃⁻) and carbonate (CO₃²⁻) in the anolyte was assessed over 1 h at 50 mA cm⁻² using the total alkalinity method to measure the crossover through the AEM. A current density of 50 mA cm⁻² is selected to investigate the ion

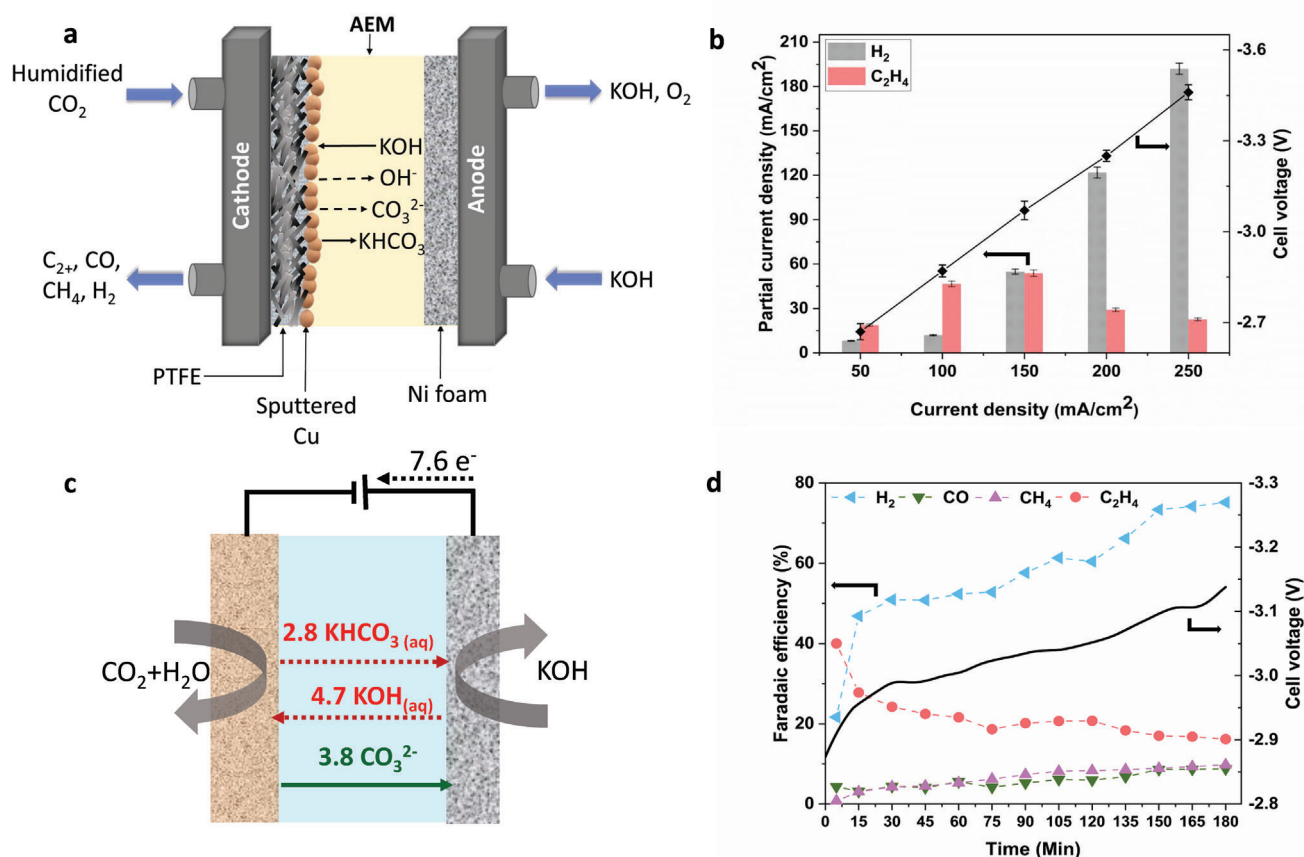


Figure 1. eCO₂R using Sustainion X37-50 Grade RT anion exchange membrane (AEM) and 1 M KOH as anolyte. a) schematic diagram of the MEA system, b) partial current density of C₂H₄ and H₂ and cell voltage at different current density, c) schematic for mass and ion transport through the membrane at a current density of 50 mA cm⁻¹,^[23] the dotted arrow and the solid arrow represent the diffusion and the electromigration of ion, respectively (see Section H, Supporting Information for details) and, d) change in Faradaic efficiency of gas products and cell voltage at 150 mA cm⁻² current density with time; using 100 mL of 1 M KOH as anolyte in (b-d).

transport throughout the study for a steady comparison and the detailed calculations are shown under Section H, Supporting Information.

It is reported that AEM can provide the most favourable alkaline reaction environment which helps to improve the activity and selectivity of eCO₂R.^[34] Also, alkaline electrolytes have been observed to substantially decrease the overpotential in eCO₂R allowing the use of non-noble Ni catalyst.^[36–40] and studies even indicate the attainment of current densities exceeding 1 A cm⁻² with the application of alkaline electrolytes.^[26,41] However, as reported in our previous work, the excessive water transport through AEM can flood the serpentine channel, accelerate salt precipitation^[42] and subsequently hinder the gas diffusion with increasing current density^[43] which will degrade the overall eCO₂R performance. Also, the (bi)carbonate crossover across AEM neutralizes the anolyte pH over time adding more overpotential to the system (Figure 1d). Moreover, membranes with high water uptake tend to experience poor mechanical strength and durability because of swelling^[44,45] which affects the overall stability of the system which is consistent with Figure 1d, where the FE_{C₂H₄} dropped by 46% and 70% in 1 and 3 h of run, respectively. These findings suggest that prolonged operation in this mode would lead to a

deterioration in both eCO₂R product selectivity and near-neutral anolyte conditions (due to CO₃²⁻ crossover), consequently causing degradation of the Ni anode.^[46] Thus, efficient ion transport and optimum water uptake through AEM is crucial to maintain the membrane hydration required for target eCO₂R and simultaneously, to address the challenge of (bi)carbonate formation and water accumulation at the cathode.^[34,35]

To resolve these challenges, thinner membranes were reported previously to reduce the water flux coming from the anode owing to elevated water repulsion by hydrophobic cathode and avoid the water accumulation at the cathode which can allow uninterrupted CO₂ diffusion through the GDL.^[23,47] As thinner membrane less than 20 μm is not commercially available, we aimed to deposit a thin layer of ionomer on top of the catalyst using a spray-coating approach (Figure 2a) (see Section C, Supporting Information for details). In comparison to a freestanding membrane, the directly deposited membrane has previously demonstrated a noteworthy drop in three primary loss mechanisms like kinetic losses, ohmic losses, and mass transport losses which translated into superior eCO₂R performance.^[23,48,49] Since the AEM with alkaline electrolyte has already showcased enhanced eCO₂R activity,^[4,6] particularly in maintaining local alkalinity which leads to

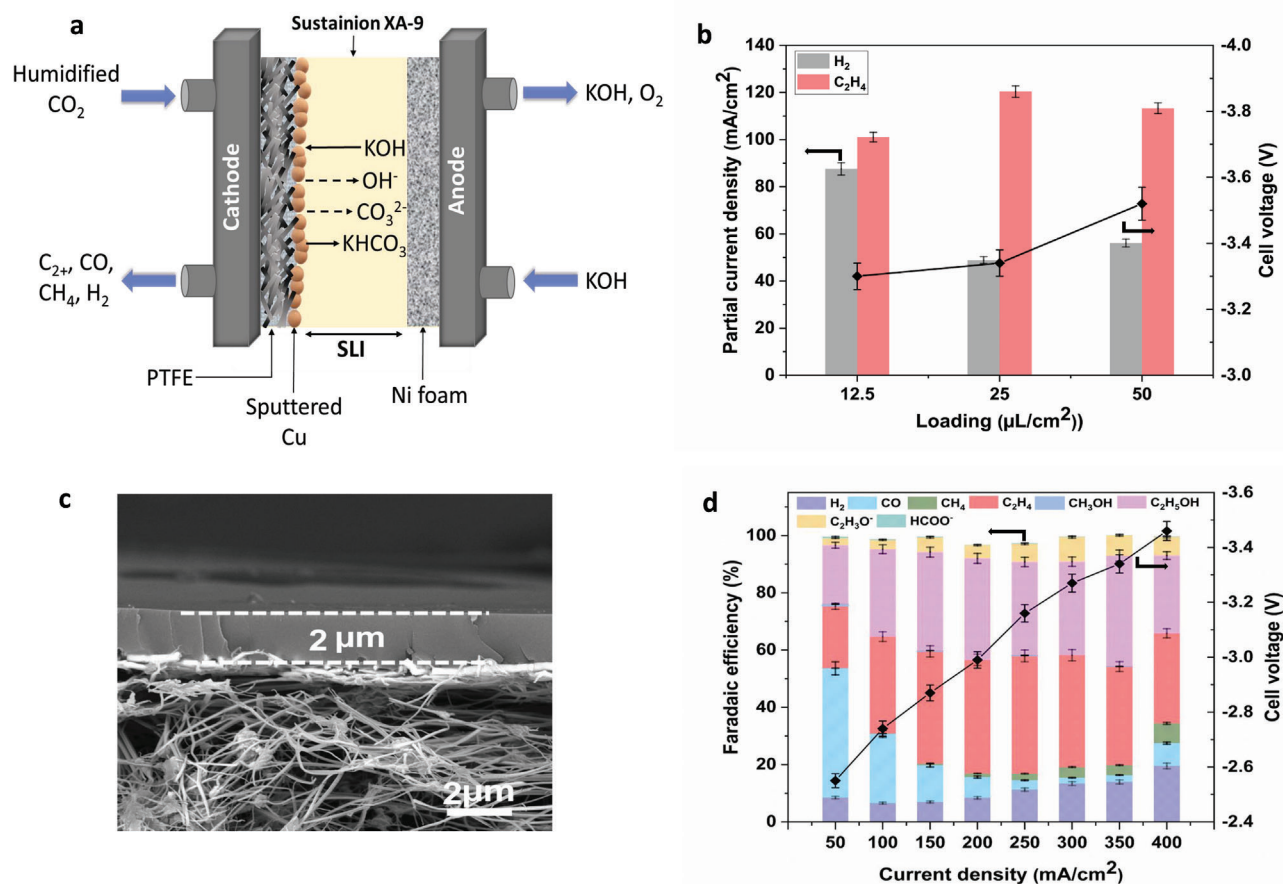


Figure 2. a) Schematic diagram of the MEA system using SLI on Cu catalyst, b) partial current density toward C₂H₄ and H₂ and cell voltage at different loadings of Sustainion XA-9 at 350 mA cm⁻², c) cross-sectional SEM image of SLI-25, d) total products distribution and cell voltage at different current density using SLI-25; using 100 mL of 1 M KOH as anolyte in (b) and (d).

substantial suppression of HER,^[4,6,50] we chose Sustainion XA-9 for this direct deposition approach in conjunction with alkaline anolyte. To airbrush the Sustainion XA-9, we utilized three different solvents (Deionized water or DI water, methanol or MeOH, and a combination of isopropyl alcohol or IPA and DI water) to ensure the proper dispersion of the ionomer solution. Among these, DI water exhibits a stronger affinity for the styrene backbones of the Sustainion XA-9 ionomer compared to other solvents and provides moderate ionomer aggregation yielding a more cohesive film.^[51,52] Thereby, we opted for DI water as the coating with MeOH and IPA/DI water was not homogenous due to poor dispersion and translated to poor electrochemical performance^[53] (Figure S4, Supporting Information).

With this airbrushing approach, there is intimate and uniform contact between the catalyst and the ionomer layer,^[49,54,55] and thus the catalyst could be protected from the commonly reported degradation mechanisms (e.g., poisoning, agglomeration, dissolution etc.) with continuous CO₂ transport for efficient eCO₂R.^[56–58] Also, the close proximity of the catalyst to the membrane or ionomer helps to minimize the ohmic and mass transfer losses.^[34,49,54] Moreover, the imidazolium group in Sustainion XA-9 has the affinity for CO₂ that can increase the local CO₂ availability, and can modulate the electric field to improve the

C₂₊ product selectivity.^[59] As mentioned earlier, a reduced thickness of the ionomer film can attenuate the water flux originating from the anode, facilitated by the back convection by the thin hydrophobic cathode. This mechanism in turn prevents the accumulation of water at the cathode GDL, thereby enhancing the availability of CO₂.^[23,47,55]

Initially, we tested different loadings (12.5, 25, and 50 μL cm⁻²) of the Sustainion XA-9 (SLI- 12.5, 25, and 50) to observe both product selectivity and their corresponding ion transports through the membrane (Figure 2b; Figures S5 and S6, Supporting Information). As shown in Figure S6, Supporting Information, the water or salt accumulation is quantified as 1.2, 0.3, and 2.4 millimole (mmol) (net potassium ion, K⁺ calculated from the transport of KOH and potassium bicarbonate, KHCO₃) for SLI- 12.5, 25, and 50, respectively which can affect the local CO₂ availability. However, numerous other factors can significantly influence the system performance, including the intricate trade-offs involving water uptake, CO₂ permeability, and the associated mass transport mechanism between electrodes. This aspect necessitates a more comprehensive and systematic understanding, indicating scopes for future studies in this field. However, observing noticeable performance enhancement with SLI-25 among the three different loadings (Figure 2b), we thereby

focused on SLI-25 for further experiments optimizing the balance between CO_2 permeability and water uptake. The scanning electron microscopy (SEM) analysis of SLI-25 using DI water solvent shows a uniform cross-section (Figure 2c) and the thickness measured from the SEM shows $\approx 2 \mu\text{m}$. The total product distribution and cell voltage have been measured at different current densities (Figure 2d). We found a maximum FE $\approx 80\%$ for C_{2+} products ($\approx 34\% \text{C}_2\text{H}_4$, $\approx 39\% \text{C}_2\text{H}_5\text{OH}$, and $\approx 7\% \text{C}_2\text{H}_3\text{O}^-$) and $\approx 14\% \text{H}_2$ at 350 mA cm^{-2} which leads to a remarkable $j_{\text{C}_{2+}}$ of $\approx 280 \text{ mA cm}^{-2}$.

We further measured the mass transport for SLI-25 (Figure S6b, Supporting Information) and since the charge transfer is governed by only unidirectional CO_3^{2-} in SLI, indicating that this configuration is acting as an AEM. We also found that with SLI-25, 14.6 mmol KOH is transported from anode to cathode while 14.3 mmol KHCO_3 is crossing from cathode to anode at 50 mA cm^{-2} in 1 h leaving only $0.3 \text{ mmol (K}^+)$ at the cathode (Figure S6b, Supporting Information). Based on this, it is calculated that K^+ accumulation at the cathode was reduced by $\approx 84\%$ compared to the AEM in Figure 1c (1.9 mmol K^+ as residue) owing to lower water uptake and back convection across the thin ionomer layer. This ultimately suppresses the water accumulation or flooding at cathode enhancing the local CO_2 availability. Later, the water uptake of the membranes was also evaluated (see Section I, Supporting Information for details) as displayed in Figure 4d where the SLI-25 shows $\approx 39\%$ reduction in water uptake compared to the AEM.

This thereby proves that thinner membranes like SLI-25 with lower water uptake can reduce the excess K^+ deposition as well as suppress cathode flooding to improve CO_2 availability. As a result, a significant enhancement in $j_{\text{C}_{2+}}$ ($\approx 280 \text{ mA cm}^{-2}$) is observed with SLI-25 compared to commercial AEM.

Behaving like an AEM, SLI also promotes KOH diffusion from anode to cathode maintaining local alkalinity near the catalyst surface which can decrease the activation energy required for CO-CO dimerization.^[60,61] As mentioned previously, it can play a favorable role by promoting eCO_2R kinetics and enhancing eCO_2R selectivity over competitive HER. However, owing to the thin layer ($\approx 2 \mu\text{m}$), SLI tends to have very high CO_2 permeability (i.e., CO_2 escapes from cathode) (see Section K; Figure S15, Supporting Information) which makes it easier to react with the incoming KOH . Thus, excessive KOH diffusion can lead to CO_2 loss due to the formation of (bi)carbonate salt (KHCO_3). We thus measured the KOH diffusion across the SLI-25 over 1 h, and found it to be 14.6 mmol (Figure S6b, Supporting Information), which is threefold higher as compared to that of AEM (Figure 1c). Over time this can cause alkalinity loss of anolyte driven by the concentration gradient of (bi)carbonate toward the anode and rapidly destabilize the Ni anode (Figure S7, Supporting Information). This indicates that the synergetic effect of ion transport and CO_2 availability (due to intimate contact of ionomer to catalyst) is required to enhance the eCO_2R stability.

As a next step, we devise a bilayer ionomer (BLI) strategy to mitigate excessive KOH diffusion and carbonate crossover in light of our recently published work using a directly deposited ultra-thin Nafion layer.^[23] Figures 3a,b show the two possible configurations to design a bilayer strategy. In the forward bias mode, in Figure 3a (i.e., anion exchange layer or AEL toward cathode and cation exchange layer or CEL toward anode; denoted as

Cu/AEL/CEL), cations (K^+) and anions (OH^- , CO_3^{2-}) from anode and cathode, respectively could easily transport to the AEL/CEL interface, resulting in salt accumulation. To confirm this notion, we deposited an additional layer of $25 \mu\text{L cm}^{-2}$ ionomer (Nafion) on top of the SLI-25 (Sustainion XA-9) to develop a BLI configuration and tested the eCO_2R performance. We found a maximum $j_{\text{C}_{2\text{H}_4}}$ of $\approx 13 \text{ mA cm}^{-2}$ for this configuration, which is fourfold lower compared to that of AEM (Figure S8a, Supporting Information, and Figure 1b). Furthermore, the configuration also suffered from poor eCO_2R stability originating from K^+ trapped at the ionomer interface and thus, accelerating excessive (bi)carbonate salt diffusion to anode as evident from rapid change in voltage in Figure 3c.

Based on this finding, we then hypothesized that a reverse bias bilayer configuration (Cu/CEL/AEL) could address the issue of salt accumulation and low eCO_2R selectivity. As shown in Figure 3b, Nafion (CEL) on top of Cu catalyst, can trap the anions generated at the cathode (OH^- , CO_3^{2-}) due to its negative background charge, thus mitigating carbonate crossover and CO_2 loss. Also, the trapped OH^- generated from eCO_2R can increase the local pH at the catalyst surface to enhance the reaction kinetics.^[24] Moreover, the trapped OH^- and CO_3^{2-} in presence of water can also promote CO_2 regeneration through series of reactions.^[62,63] On the other hand, Sustainion XA-9 (AEL) with a positive background charge, can inhibit K^+ migration toward cathode which reduces the chances of salt accumulation at the cathode. Thus, we decided to choose the final BLI stacking order as Cu/CEL/AEL or $\text{Cu/Nafion/Sustainion XA-9}$ as shown in schematic in Figure 3d. The SEM image in Figure 3e (and energy dispersive x-ray spectroscopy or EDS mapping under Figure S9, Supporting Information) shows two immiscible layers of ionomers with thickness of ≈ 1 and $\approx 2 \mu\text{m}$ for $25 \mu\text{L cm}^{-2}$ of Nafion and $25 \mu\text{L cm}^{-2}$ Sustainion XA-9, respectively (denoted as BLI-25:25). As expected, BLI-25:25 in reverse bias configuration was more stable (Figure 3f) than the forward bias mode ($\approx 6 \text{ h}$ vs $\approx 1 \text{ h}$ respectively) due to reduced salt accumulation (Figure 3b). However, while this reverse bias configuration slightly improved the eCO_2R activity ($j_{\text{C}_{2\text{H}_4}}$ of $\approx 22 \text{ mA cm}^{-2}$) (Figure S8b, Supporting Information) compared to the forward bias stacking order ($j_{\text{C}_{2\text{H}_4}}$ of $\approx 13 \text{ mA cm}^{-2}$), the $j_{\text{C}_{2\text{H}_4}}$ was still low (Figure S5b, Supporting Information) compared to the SLI-25. To assess the inherent reasons behind such poor selectivity but stable eCO_2R , the corresponding ion transport across the reverse biased BLI-25:25 was assessed showing very low diffusion of KOH (Figure S10, Supporting Information). We thus hypothesized that the resulting OH^- diffusion was too low (0.2 mmol in 1 h as shown in Figure S10 (Supporting Information) which is ≈ 73 -fold lower than SLI-25 in Figure S6b, Supporting Information) to provide an alkaline micro-environment required for the C_{2+} reaction pathways.^[43] In the absence of enough OH^- , the $-\text{SO}_3\text{H}$ groups of Nafion which is in close contact with Cu can acidify the surface to some extent promoting HER in turn.^[23]

We also observed that the hydrated K^+ deposition at the cathode has been reduced significantly (Figure S10, Supporting Information) in BLI-25:25 which means that the cathode is less flooded as compared to the AEM (by $\approx 68\%$), which explains the extended performance of the reverse bias operation (Figure 3f). However, recent studies have also demonstrated a threefold

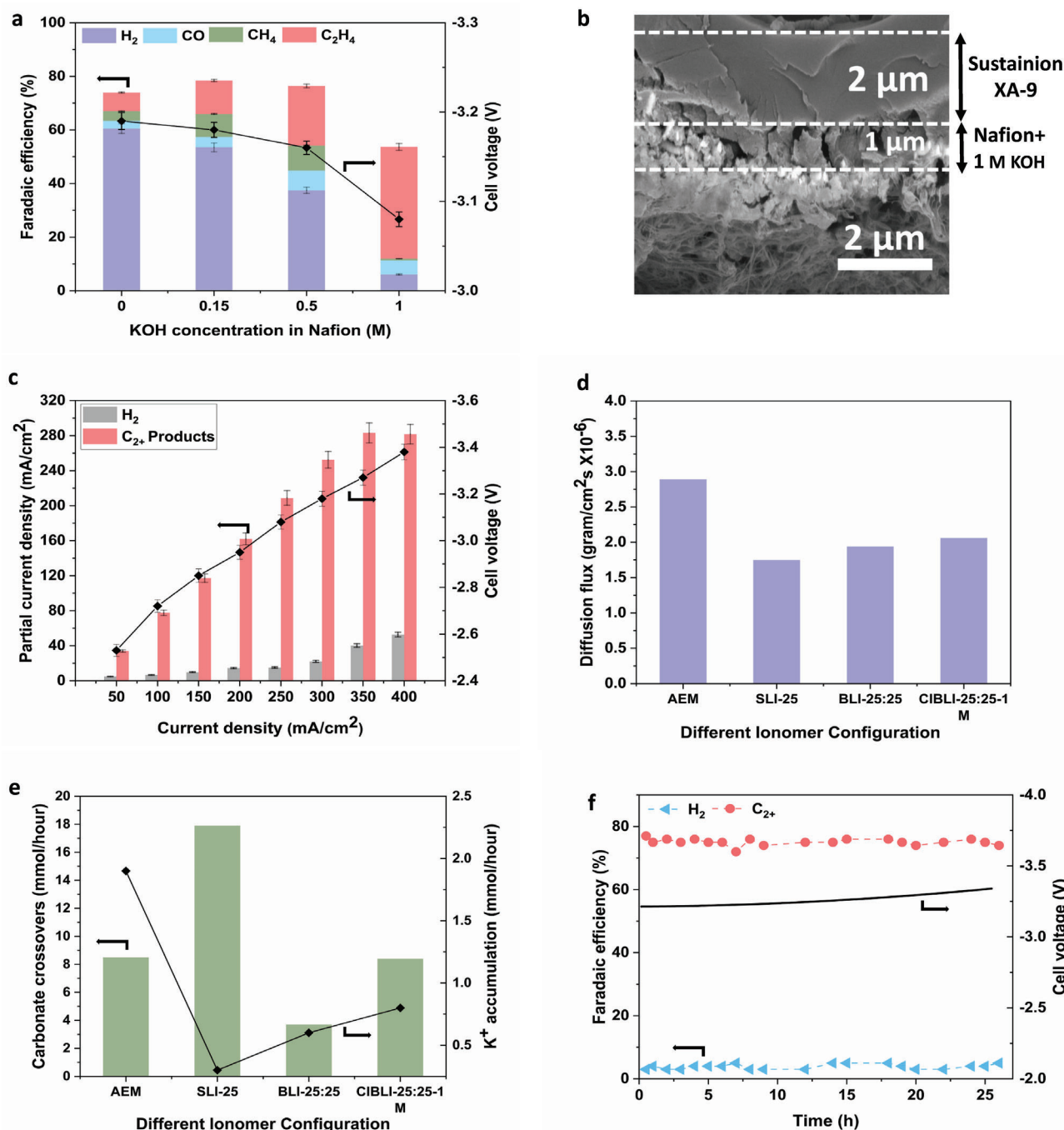


Figure 4. Electrochemical performance and structure for CIBLI-25:25-1 m; a) gas products distribution and cell voltage at 250 mA cm⁻² using different concentration of KOH infusion, b) cross-sectional SEM image; c) partial current density toward C₂₊ and H₂ at different current density; using 100 mL of 1 M KOH as anolyte in (a) and (c), d) measured water diffusion flux across the commercial AEM and different ionomer configurations; using initial anolyte volume 100 mL running the experiments for 24 h under no applied potential, e) total carbonate crossover and K⁺ accumulation at the cathode while using commercial AEM and different ionomer configurations; at 50 mA cm⁻² current density using 100 mL of 1 M KOH as anolyte over 1 h duration of time (f) change in cell voltage and Faradaic efficiency of C₂₊ and H₂ at 350 mA cm⁻² over 24 h period of time.

increase in eCO₂R selectivity owing to the migration of K⁺ from the anode to the cathode.^[64] In light of this, we concluded that the lower K⁺ electromigration enables system stability, yet a minimum amount of OH⁻ or K⁺ diffusion is critical for membrane hydration and selective eCO₂R.^[23]

The key benefit to utilize Nafion is its ability to allow cation migration and block CO₃²⁻ crossover. However, excess cation (like K⁺, H⁺) migration is detrimental to eCO₂R performance. Moreover, as the eCO₂R reaction pathways are pH sensitive, it is also necessary to neutralize the acidic nature of the Nafion,

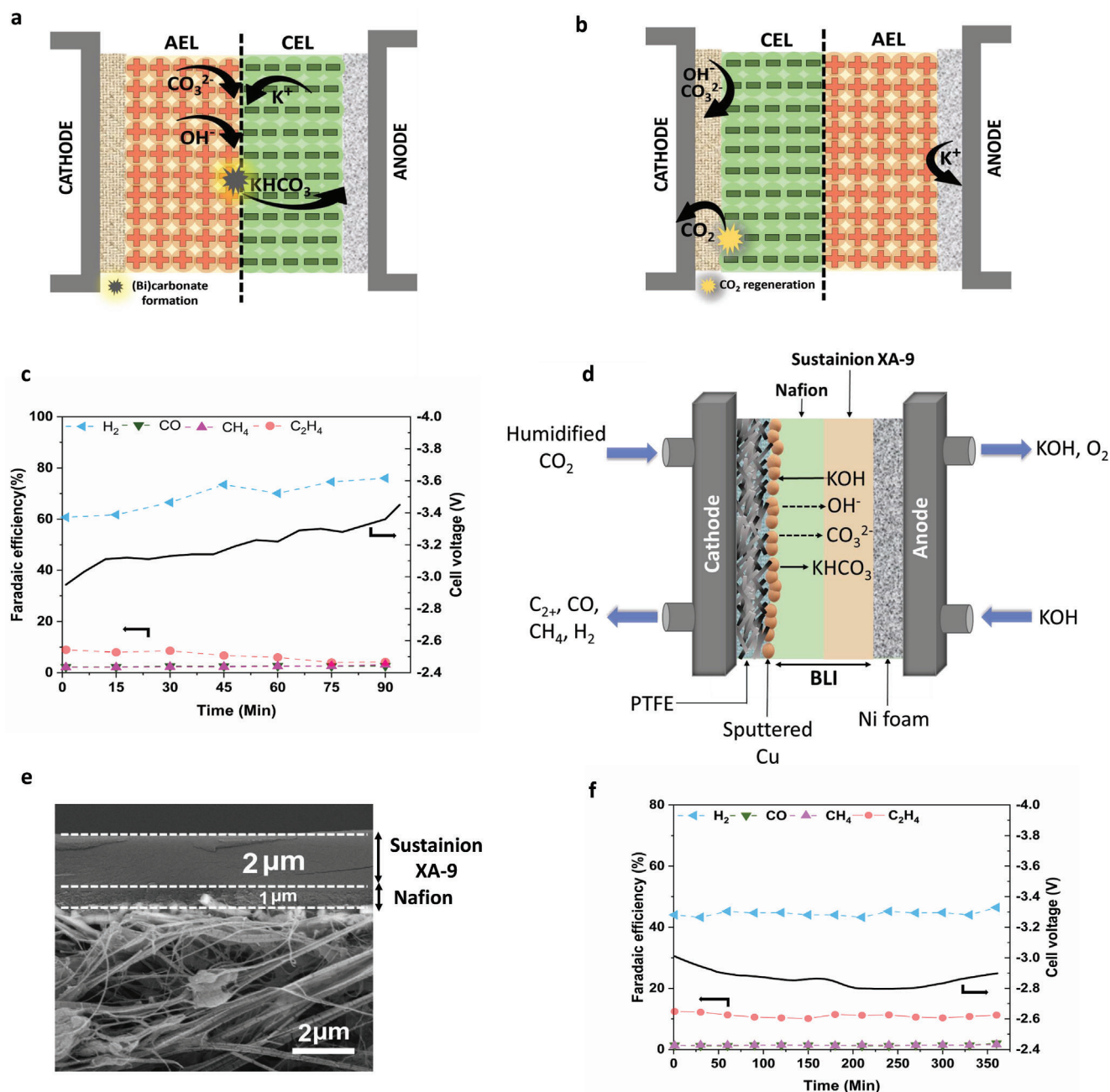


Figure 3. The design strategy of bilayer ionomer configurations operating in (a) forward bias mode (Cu/AEL/CEL), (b) reverse bias mode (Cu/CEL/AEL), (c) change in Faradaic efficiency of gas products and cell voltage with time at 150 mA cm⁻² using forward bias mode, (d) schematic diagram of the MEA system using BLI-25:25 on Cu catalyst in reverse bias mode, (e) cross-sectional SEM image of BLI-25:25, (f) change in Faradaic efficiency of gas products and cell voltage with time at 150 mA cm⁻² in reverse bias mode; using 100 mL of 1 M KOH as electrolyte in (c) and (f).

which is closest to the catalyst layer to suppress HER. One such strategy is to replace the H^+ groups of sulphonic acid in Nafion with larger cations (e.g., K^+ , Na^+ , Cs^+) to suppress the ionic conductivity (lower K^+ migration) as well as acidity (lower H^+ concentration).^[60,61] Our previous study also reported that the implanted cations on the Nafion enhance eCO_2R performance, while the bulk cations transported from the anode to cathode have a detrimental effect as solvated cations carry water molecules along.^[34] Therefore, we hypothesized that cation

infused bilayer would create an optimal alkaline local environment due to tuning of H^+ amount and can maintain sufficient hydration for membrane conductivity without the accumulation of excess K^+ and water on cathode surface.

To analyze this hypothesis, we tested several cation-infused bilayer ionomer (CIBLI) with three different loadings of Nafion (12.5, 25, and 50 μL) (Figure S11, Supporting Information) where Cu/25 μL Nafion/25 μL Sustainion XA-9 (CIBLI-25:25) showed the maximum selectivity and least overpotential. Also, different

concentrations of cation infusion were scrutinized with the optimized ionomer loading by mixing KOH with Nafion solution (by varying KOH concentration from 0 to 1 M). The selectivity and the voltage of CIBLI-25:25 improved gradually with the increase in KOH concentration, and the maximum selectivity was obtained with 1 M KOH infusion (Figure 4a). The final optimized cation implanted bilayer system thus becomes as Cu/25 μ L Nafion with 1 M KOH infusion/25 μ L Sustainion XA-9 (denoted as CIBLI-25:25-1 M). The thickness of the optimized sample under SEM (Figure 4b) was ≈ 1 μ m and ≈ 2 μ m, respectively for the inner (Nafion+1 M KOH) and outer (Sustainion XA-9) layers, respectively, which is identical to the one in the absence of cation in Figure 3e. Additionally, the top view SEM (Figure S12, Supporting Information) confirms the homogenous distribution of the ionomers on the Cu catalyst.

Later, we investigated the FE and j for CIBLI-25:25-1 M at different current densities as shown in Figure 4c. The maximum $j_{C_{2+}}$ achieved was ≈ 284 mA cm $^{-2}$ at 350 mA cm $^{-2}$ with $\approx 81\%$ FE $_{C_{2+}}$ ($\approx 41\%$ C $_2$ H $_4$, $\approx 35\%$ C $_2$ H $_5$ OH and $\approx 5\%$ C $_2$ H $_3$ O $^-$) and $\approx 12\%$ FE $_{H_2}$ (Figure 4c; Figure S13, Supporting Information). According to the literatures,^[46,65–68] the alkali cations help in cathode activation promoting the C–C bond formation and thus resulting in selective C $_{2+}$ products. Also, those findings revealed that the presence of certain local electric fields induced by cations has a significant stabilizing effect on adsorbed CO $_2$ molecules making it much easier to undergo eCO $_2$ R.^[65,67,68] To correlate to the aforesaid performance, we further measured the mass transport across CIBLI-25:25-1 M over 1 h (see Section H, Supporting Information for details). The electromigration is governed by CO $_3^{2-}$ and the diffusive transport for KOH and KHCO $_3$ is primarily dictated by the membrane's water uptake property as evident in Figure 4d. We found that the water uptake of CIBLI-25:25-1 M is $\approx 29\%$ lower than that of AEM, as evident from the water flux measurement (Figure 4d). Although transport of anions like HCO $_3^-$, CO $_3^{2-}$ can occur in both aqueous potassium carbonate (K $_2$ CO $_3$) and KHCO $_3$ form over time but for simplicity we assumed that all the anions diffuse as KHCO $_3$ only. The results show that the (bi)carbonate diffusion or crossover (Figure S14, Supporting Information) for CIBLI-25:25-1 M is similar to that of AEM (Figure 1c). Still the water accumulation and salt precipitation at the cathode are suppressed by $\approx 58\%$ (solvated K $^+$) compared to the AEM (Figure 4e; Figure S15, Supporting Information) which in turn results in high CO $_2$ availability at the catalyst vicinity required for efficient eCO $_2$ R. However, this value is highly dependent on specific reaction parameters and operating conditions and can vary with changing electrolyte concentration, current density, membrane thickness etc.

Also, the lower voltage observed in the directly deposited membrane configurations (i.e., 2.95 V with CIBLI-25:25-1 M in Figure 4c vs 3.25 V with AEM in Figure 1b at 200 mA cm $^{-2}$ current density) can be attributed to the ultra-thin layer and improved ionic conductivity.^[69] Electrochemical Impedance Spectroscopy (EIS) studies were carried out to support this idea (see Section J, Supporting Information for details). From the Figure S16a, Supporting Information, the commercial AEM displays the highest R_s value of 8.37 Ω which is due to the thicker membrane interface which increases ohmic resistance. When the SLI is tested for EIS, as expected, the R_s value dropped to 4.46 Ω im-

plying the influential role of the directly deposited layer at the catalyst surface in decreasing the cell resistance. In the case of BLI, a slightly higher resistance of 5.13 Ω was found, which can be well correlated to its higher thickness. The addition of 1 M KOH to Nafion (CIBLI-25:25-1 M) improves the wetting of the catalyst surface and shows a lower R_s of 4.94 Ω . The bar diagram provided in Figure S16b, Supporting Information portrays the clear differences between the commercial and manually air-brushed membranes on the catalyst surface. From the R_s values observed, the lower resistance of the CIBLI-25:25-1 M system can directly be correlated to its high ionic conductivity, thus demonstrating the superiority of CIBLI over commercial AEM in eCO $_2$ R.^[42]

As mentioned earlier, we believe that high (bi)carbonate formation in the SLI compared to BLI and CIBLI can be attributed to the high CO $_2$ permeability. To verify this, we have designed an experiment to measure the CO $_2$ flux across the directly deposited membranes (see Section K, Supporting Information for details). From this analysis, we confirmed our assumption of high CO $_2$ permeability with SLI-25. We discovered that the CO $_2$ permeability is indeed very high and thus a significant amount of CO $_2$ is lost and a part of it reacts chemically with KOH to form KHCO $_3$. Thus, the additional Nafion layer near the catalyst vicinity in BLI and CIBLI configuration is aiding to reduce the gas permeability and hereby mitigating CO $_2$ loss and KHCO $_3$ formation (Figure S17, Supporting Information).

Finally, we carried out the extended eCO $_2$ R experiment to understand the stability of CIBLI-25:25-1 M in a MEA configuration. The CIBLI configuration suppressed K $^+$ crossover to the cathode due to electromigration as well as water diffusion. The CIBLI further enhanced C $_{2+}$ selectivity at higher current densities due to high CO $_2$ availability at the reaction vicinity. We demonstrated over 24 h of stable operation using CIBLI-25:25-1 M at the cathode and Ni at the anode; using 1 M KOH anolyte with the CO $_2$ -to-C $_{2+}$ electrolyzer efficiency of 29% at 350 mA cm $^{-2}$. Also, it is understood that optimizing the concentration of product gas is crucial for minimizing costs associated with the gas separation of diluted product streams. In our investigation, we systematically reduced the flow rate of CO $_2$ from 50 to 10 sccm for the CIBLI-25:25-1 M. The impact of CO $_2$ depletion along the gas channel was monitored and the selectivity could be retained up to 15 sccm (Figure S18, Supporting Information). This observation along with higher $j_{C_{2+}}$ leads us to identify a single pass conversion efficiency of 23% and a minimal requirement of electrolyzer conversion energy of 117 GJ per ton of C $_{2+}$ products with electrolyzer energy efficiency (EE) of 29% toward C $_{2+}$ products in one step eCO $_2$ R (see Section M; Table S.3, Supporting Information for details).

3. Conclusion

In conclusion, we demonstrated the potential of Sustainion XA-9 and Nafion ionomers in modulating the reaction environment to achieve remarkable $j_{C_{2+}}$ while preserving selectivity and stability. The ultra-thin single-layer Sustainion XA-9 suppresses the cathode flooding due to water repulsion by hydrophobic PTFE-based gas diffusion layer. Sustainion ionomer, owing to its high CO $_2$ solubility and alkaline nature, initially exhibited an impressive $j_{C_{2+}}$ of ≈ 280 mA cm $^{-2}$. However, it also posed challenges

related to (bi)carbonate formation and CO₂ loss. Introducing an additional Nafion ionomer layer in reverse bias mode helped us to suppress the (bi)carbonate crossover significantly by controlling OH[−] and K⁺ transport. Later, incorporating K⁺ ions into the innermost Nafion layer, we facilitated C—C bond formation leading toward high C₂₊ selectivity. Overall, this ultra-thin (≈3 μm) bi-layer configuration (CIBLI-25:25-1 m) reduced the cathode flooding or salt accumulation by ≈58% compared to commercially available AEM permitting high CO₂ near catalyst vicinity. This also achieved a comparable $j_{C_{2+}}$ of ≈284 mA cm^{−2} (≈81% selectivity toward C₂₊ products at 350 mA cm^{−2} current density) with a stability over 24 h. This facile, cost-effective, and scalable approach of directly deposited ultra-thin polymer combination completely bypasses the need for a commonly used prefabricated membrane along with its associated cost of production, pretreatment, hot pressing, assembling, etc. Additionally, this system eliminates the necessity of using commonly used expensive metal anode (e.g., IrO₂) and allows eCO₂R with alkaline anolyte offering a minimal conversion energy of 117 GJ/ton C₂₊ products with an electrolyzer energy efficiency (EE) of 29% at 350 mA cm^{−2} current density in one-step CO₂ conversion towards C₂₊ products.

Overall, this study underscores the potential of tailored ionomer design and layering strategies in shaping the electrochemical environment. The study highlights the intricate interplay of factors that influence eCO₂R performance, including water diffusion and salt accumulation (to minimize flooding), CO₂ availability (to increase C₂₊ selectivity), and local pH control (to provide alkaline microenvironment). However, the impact of specific reaction parameters such as electrolyte concentration, current density, membrane thickness etc. still needs to be fully understood on the performance of the bilayer ionomer configuration. Also, future advancements in this field can be directed toward tailoring ionomer properties, exploring new ionomer materials, optimizing the layering configuration, and performing operando/in situ characterizations to achieve even higher $j_{C_{2+}}$ while minimizing undesired side reactions. Additionally, the insights gained from this study can be utilized in the design and development of other efficient electrochemical systems for sustainable applications.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors gratefully acknowledge the financial support provided by the Canada First Research Excellence Fund (CFREF) at the University of Calgary for funding this project.

Conflict of Interest

M.G.K., M.A.A., and M.R. have submitted Invention Disclosure Application WIPO (PCT) Patent Application No. CA2023/051424 with a title “ELECTRODES USEFUL FOR MEMBRANE-FREE ELECTRODE ASSEMBLIES, MEMBRANE-FREE ELECTRODE ASSEMBLIES, METHODS, AND USES THEREOF”.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

cathode flooding, directly deposited membrane, ionomers, Partial current density, Sustainion XA-9

Received: February 3, 2024

Revised: July 2, 2024

Published online:

- [1] S. Nitopi, E. Bertheussen, S. B. Scott, X. Liu, A. K. Engstfeld, S. Horch, B. Seger, I. E. L. Stephens, K. Chan, C. Hahn, J. K. Nørskov, T. F. Jaramillo, I. Chorkendorff, *Chem. Rev.* **2019**, *119*, 7610.
- [2] P. De Luna, C. Hahn, D. Higgins, S. A. Jaffer, T. F. Jaramillo, E. H. Sargent, *Science* (80-). **2019**, *364*, 6438.
- [3] M. Jouny, W. Luc, F. Jiao, *Ind. Eng. Chem. Res.* **2018**, *57*, 2165.
- [4] M. G. Kibria, J. P. Edwards, C. M. Gabardo, C. T. Dinh, A. Seifitokaldani, D. Sinton, E. H. Sargent, *Adv. Mater.* **2019**, *31*, 1807166.
- [5] S. Verma, B. Kim, H. R. M. Jhong, S. Ma, P. J. A. Kenis, *ChemSusChem* **2016**, *9*, 1972.
- [6] G. O. Larrazábal, M. Ma, B. Seger, *Accounts Mater. Res.* **2021**, *2*, 220.
- [7] J. A. Rabinowitz, M. W. Kanan, *Nat. Commun.* **2020**, *11*, 10.
- [8] M. Ma, S. Kim, I. Chorkendorff, B. Seger, *Chem. Sci.* **2020**, *11*, 8854.
- [9] K. J. Puring, D. Siegmund, J. Timm, F. Möllenbruck, S. Schemme, R. Marschall, U. Apfel, *Adv. Sustain. Syst.* **2021**, *5*, 2000088.
- [10] Y. Hori, A. Murata, R. Takahashi, *J. Chem. Soc. Faraday Trans. 1 Phys. Chem. Condens. Phases* **1989**, *85*, 2309.
- [11] R. Kas, R. Kortlever, A. Milbrat, M. T. M. Koper, G. Mul, J. Baltrusaitis, *Phys. Chem.* **2014**, *16*, 12194.
- [12] A. S. Varela, M. Kroschel, T. Reier, P. Strasser, *Catal. Today* **2015**, *260*, 8.
- [13] X. Chen, Y. Zhao, J. Han, Y. Bu, *ChemPlusChem* **2023**, *88*, 202200370.
- [14] N. Tran, H. P. Duong, S. Zanna, M. W. Schreiber, M. Fontecave, *ACS Appl. Mater. Interfaces* **2022**, *14*, 31933.
- [15] Y. Yang, S. Louisia, S. Yu, J. Jin, I. Roh, C. Chen, M. V. Fonseca Guzman, J. Feijóo, P. C. Chen, H. Wang, C. J. Pollock, X. Huang, Y. T. Shao, C. Wang, D. A. Muller, H. D. Abruña, P. Yang, *Nature* **2023**, *614*, 262.
- [16] S. Nitopi, E. Bertheussen, S. B. Scott, X. Liu, A. K. Engstfeld, S. Horch, B. Seger, I. E. L. Stephens, K. Chan, C. Hahn, J. K. Nørskov, T. F. Jaramillo, I. Chorkendorff, *Chem. Rev.* **2019**, *119*, 7610.
- [17] X. Xie, L. Du, L. Yan, S. Park, Y. Qiu, J. Sokolowski, W. Wang, Y. Shao, *Adv. Funct. Mater.* **2022**, *32*, 2110036.
- [18] C. Hu, L. Zhang, J. Gong, *Energy Environ. Sci.* **2019**, *12*, 2620.
- [19] T. Luo, S. Abdu, M. Wessling, *J. Memb. Sci.* **2018**, *555*, 429.
- [20] Z. Zhang, Q. Lu, J. Sun, G. Li, W. Wu, Z. Xu, L. Xu, Y. Wang, *Chem. Sci.* **2024**, *15*, 2786.
- [21] G. O. Larrazábal, P. Strøm-Hansen, J. P. Heli, K. Zeiter, K. T. Therkildsen, I. Chorkendorff, B. Seger, *ACS Appl. Mater. Interfaces* **2019**, *11*, 41281.
- [22] T. Lee, Y. Lee, J. Eo, D. H. Nam, *Nanoscale* **2023**, *16*, 2235.
- [23] M. A. Adnan, A. S. Zeraati, S. K. Nabil, T. A. Al-attas, K. Kannimuthu, C. Dinh, I. D. Gates, G. Kibria, *Adv. Energy Mater.* **2023**, *13*, 2203158.
- [24] C. Kim, J. C. Bui, X. Luo, J. K. Cooper, A. Kusoglu, A. Z. Weber, A. T. Bell, *Nat. Energy* **2021**, *6*, 1026.

- [25] T. Alkayyali, A. S. Zeraati, H. Mar, F. Arabyarmohammadi, S. Saber, R. K. Miao, C. P. O. Brien, H. Liu, Z. Xie, G. Wang, E. H. Sargent, N. Zhao, D. Sinton, *ACS Energy Lett.* **2023**, *8*, 4674.
- [26] F. P. G. De Arquer, C. Dinh, A. Ozden, J. Wicks, C. McCallum, A. R. Kirmani, D. Nam, C. Gabardo, A. Seifitokaldani, X. Wang, Y. C. Li, F. Li, J. Edwards, L. J. Richter, S. J. Thorpe, D. Sinton, E. H. Sargent, *Science* **2020**, *2*, 661.
- [27] J. J. Kaczur, H. Yang, Z. Liu, S. D. Sajjad, R. I. Masel, *Front. Chem.* **2018**, *6*, 263.
- [28] Z. Yin, H. Peng, X. Wei, H. Zhou, J. Gong, M. Huai, L. Xiao, G. Wang, J. Lu, L. Zhuang, *Energy Environ. Sci.* **2019**, *12*, 2455.
- [29] Y. Xu, J. P. Edwards, J. Zhong, C. P. O'Brien, C. M. Gabardo, C. McCallum, J. Li, C. T. Dinh, E. H. Sargent, D. Sinton, *Energy Environ. Sci.* **2020**, *13*, 554.
- [30] J. J. Kaczur, H. Yang, Z. Liu, S. D. Sajjad, R. I. Masel, *Front. Chem.* **2018**, *6*, 1.
- [31] C. M. Gabardo, P. Colin, O. Brien, P. Jonathan, J. Li, H. Edward, D. Sinton, C. M. Gabardo, C. P. O. Brien, J. P. Edwards, C. McCallum, Y. Xu, C. Dinh, J. Li, E. H. Sargent, D. Sinton, *Joule* **2019**, *3*, 2777.
- [32] Y. C. Li, D. Zhou, Z. Yan, R. H. Gonçalves, D. A. Salvatore, C. P. Berlinguette, T. E. Mallouk, *ACS Energy Lett.* **2016**, *1*, 1149.
- [33] Z. Zhang, X. Huang, Z. Chen, J. Zhu, B. Endr, C. Janáky, *Angew. Chem.* **2023**, *135*, e202302789.
- [34] S. Garg, C. A. Giron Rodriguez, T. E. Rufford, J. R. Varcoe, B. Seger, *Energy Environ. Sci.* **2022**, *15*, 4440.
- [35] E. W. Lees, B. A. W. Mowbray, F. G. L. Parlance, C. P. Berlinguette, *Nat. Rev. Mater.* **2022**, *7*, 55.
- [36] S. Ponnuram, C. M. Yun, I. V. Chernyshova, *ChemElectroChem* **2016**, *3*, 74.
- [37] J. J. Lv, M. Jouny, W. Luc, W. Zhu, J. J. Zhu, F. Jiao, *Adv. Mater.* **2018**, *30*, 1703663.
- [38] M. Wang, K. Torbensen, D. Salvatore, S. Ren, D. Joulié, F. Dumoulin, D. Mendoza, B. Lassalle-Kaiser, U. Işci, C. P. Berlinguette, M. Robert, *Nat. Commun.* **2019**, *10*, 3602.
- [39] D. A. Salvatore, D. M. Weekes, J. He, K. E. Dettelbach, Y. C. Li, T. E. Mallouk, C. P. Berlinguette, *ACS Energy Lett.* **2018**, *3*, 149.
- [40] S. Verma, Y. Hamasaki, C. Kim, W. Huang, S. Lu, H. R. M. Jhong, A. A. Gewirth, T. Fujigaya, N. Nakashima, P. J. A. Kenis, *ACS Energy Lett.* **2018**, *3*, 193.
- [41] W. Ma, S. Xie, T. Liu, Q. Fan, J. Ye, F. Sun, Z. Jiang, Q. Zhang, J. Cheng, Y. Wang, *Nat. Catal.* **2020**, *3*, 478.
- [42] M. A. Adnan, A. Shayesteh Zeraati, S. K. Nabil, T. A. Al-Attas, K. Kannimuthu, C. T. Dinh, I. D. Gates, M. G. Kibria, *Adv. Energy Mater.* **2023**, *13*, 2203158.
- [43] D. G. Wheeler, B. A. W. Mowbray, A. Reyes, F. Habibzadeh, J. He, C. P. Berlinguette, *Energy Environ. Sci.* **2020**, *13*, 5126.
- [44] J. Han, L. Zhu, J. Pan, T. J. Zimudzi, Y. Wang, Y. Peng, M. A. Hickner, L. Zhuang, *Macromolecules* **2017**, *50*, 3323.
- [45] J. Pan, S. Lu, Y. Li, A. Huang, L. Zhuang, J. Lu, *Adv. Funct. Mater.* **2010**, *20*, 312.
- [46] B. Endrődi, A. Samu, E. Kecsenovity, T. Halmágyi, D. Sebők, C. Janáky, *Nat. Energy* **2021**, *6*, 439.
- [47] A. Reyes, R. P. Jansonius, B. A. W. Mowbray, Y. Cao, D. G. Wheeler, J. Chau, D. J. Dvorak, C. P. Berlinguette, *ACS Energy Lett.* **2020**, *5*, 1612.
- [48] T. Alkayyali, A. S. Zeraati, H. Mar, F. Arabyarmohammadi, S. Saber, R. K. Miao, C. P. O. Brien, H. Liu, Z. Xie, G. Wang, E. H. Sargent, N. Zhao, D. Sinton, *ACS Energy Lett.* **2023**, *8*, 4674.
- [49] P. Holzapfel, M. Bühler, C. Van Pham, F. Hegge, T. Böhm, D. McLaughlin, M. Breitwieser, S. Thiele, *Electrochem. Commun.* **2020**, *110*, 106640.
- [50] A. R. Woldu, Z. Huang, P. Zhao, L. Hu, D. Astruc, *Coord. Chem. Rev.* **2022**, *454*, 214340.
- [51] B. A. W. Mowbray, D. J. Dvorak, N. Taherimakhsoosi, C. P. Berlinguette, *Energy Fuels* **2021**, *35*, 19178.
- [52] P. Ding, H. An, P. Zellner, T. Guan, J. Gao, P. Mu, B. M. Weckhuysen, W. Van Der Stam, I. D. Sharp, *ACS Catal.* **2023**, *13*, 5336.
- [53] M. N. Idros, Y. Wu, T. Duignan, M. Li, H. Cartmill, I. Maglaya, T. Burdyny, G. Wang, T. E. Rufford, *ACS Appl. Mater. Interfaces* **2023**, *15*, 52461.
- [54] M. Klingele, B. Britton, M. Breitwieser, S. Vierrath, R. Zengerle, S. Holdcroft, S. Thiele, *Electrochem. Commun.* **2016**, *70*, 65.
- [55] P. Veh, B. Britton, S. Holdcroft, R. Zengerle, S. Vierrath, M. Breitwieser, *RSC Adv.* **2020**, *10*, 8645.
- [56] J. Huang, N. Hörmann, E. Oveisi, A. Louidice, G. L. De Gregorio, O. Andreussi, N. Marzari, R. Buonsanti, *Nat. Commun.* **2018**, *9*, 5154.
- [57] S. R. Hui, N. Shaigan, V. Neburchilov, L. Zhang, K. Malek, *Nanomaterials* **2020**, *10*, 1884.
- [58] C. A. Moreno-camacho, J. R. Montoya-torres, A. Jaegler, *J. Clean. Prod.* **2019**, *231*, 600.
- [59] M. Chang, W. Ren, W. Ni, S. Lee, X. Hu, *ChemSusChem* **2023**, *16*, 202201687.
- [60] C. T. Dinh, T. Burdyny, G. Kibria, A. Seifitokaldani, C. M. Gabardo, F. Pelayo García De Arquer, A. Kiani, J. P. Edwards, P. De Luna, O. S. Bushuyev, C. Zou, R. Quintero-Bermudez, Y. Pang, D. Sinton, E. H. Sargent, *Science (80-.)* **2018**, *360*, 783.
- [61] C. M. Gabardo, A. Seifitokaldani, J. P. Edwards, T. Cao-Thang Dinh, D. S. Burdyny, M. Golam Kibria, C. P. O'Brien, E. H. Sargent, *Energy Environ. Sci.* **2018**, *11*, 2531.
- [62] T. Li, E. W. Lees, M. Goldman, D. A. Salvatore, D. M. Weekes, C. P. Berlinguette, *Joule* **2019**, *3*, 1487.
- [63] Y. C. Xiao, C. M. Gabardo, S. Liu, G. Lee, Y. Zhao, C. P. O'Brien, R. K. Miao, Y. Xu, J. P. Edwards, M. Fan, J. E. Huang, J. Li, P. Papangelakis, T. Alkayyali, A. Sedighian Rasouli, J. Zhang, E. H. Sargent, D. Sinton, *EES Catal.* **2023**, *1*, 54.
- [64] K. Yang, M. Li, S. Subramanian, M. A. Blommaert, W. A. Smith, T. Burdyny, *ACS Energy Lett.* **2021**, *6*, 4291.
- [65] J. Resasco, L. D. Chen, E. Clark, C. Tsai, C. Hahn, T. F. Jaramillo, K. Chan, A. T. Bell, *J. Am. Chem. Soc.* **2017**, *139*, 11277.
- [66] J. H. Kim, H. Jang, G. Bak, W. Choi, H. Yun, E. Lee, D. Kim, J. Kim, Y. Lee, *Energy Environ. Sci.* **2022**, *15*, 4301.
- [67] L. D. Chen, M. Urushihara, K. Chan, *ACS Catal.* **2016**, *6*, 7133.
- [68] S. Ringe, K. Chan, S. Ringe, B. Seger, *Energy Environ. Sci.* **2019**, *12*, 3001.
- [69] H. Zhang, D. Chen, Y. Xianze, S. Yin, *Fuel Cells* **2015**, *6*, 761.